AMOLED bench shield

File: bench_display.kicad_sch

dial-me-spinner rev B bench: AMOLED shield for nRF52840-DK -- generated by hardware/design/build.py from hardware/design/sheets/bench_display.py. Do not hand-edit.

dial-me-spinner

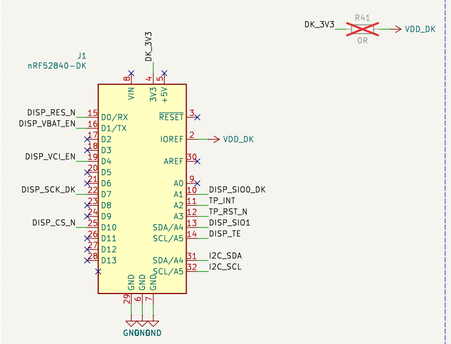
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File: dial-me-spinner-bench_display.kicad_sch

Title: dial-me-spinner rev B bench: AMOLED shield for nRF52840-DK

Size: A3	Date: 2026-10-23	**Rev: 1**
KiCad E.D.A. 9.0.9		Id: 1/2

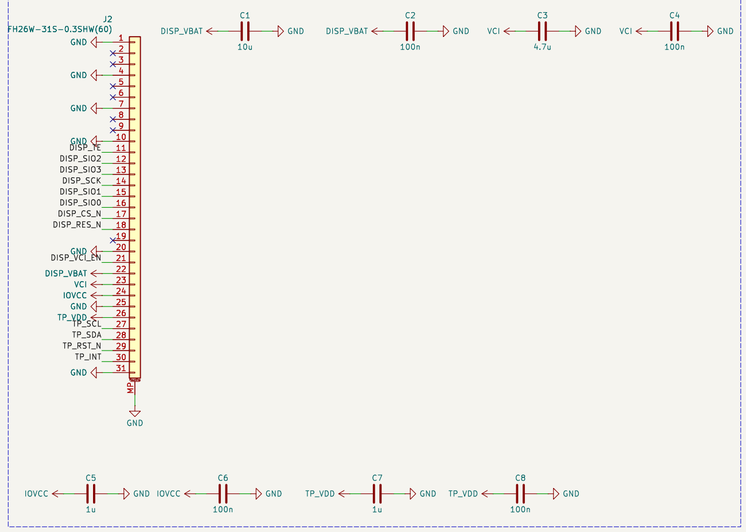
nRF52840-DK Arduino R3 headers

Pins assigned by symbol pin NUMBER: A4/SDA and A5/SCL are separate GPIOs on the DK.
32 MHz lines on the only full-speed header pins: SCK P1.08 (D7), SIO0 P0.04 (A1).
Firmware: own pinctl for SPIM3 (not arduino_spi), high drive (HDH1) on SCK and SIO0.
Logic runs at the DK's VDD (3.0 V as shipped): inside CO5300 IOVCC 1.65–3.3 V and VCI 2.4–3.3 V.



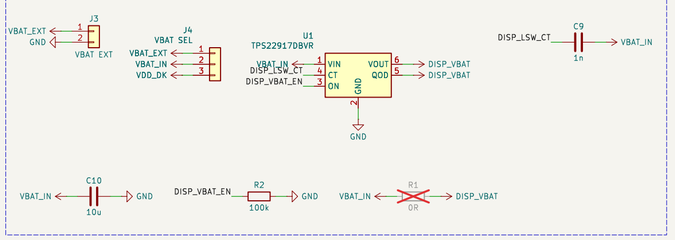
Panel connector (FH26W-31S, connector pin = 32 − panel pin)

Silk marks PANEL PIN 1 at connector pin 31. Bottom contact: FPC contacts face the board.
MTP_PWR (panel 13) open: MIPI pairs (panel 23–24, 26–27, 29–30) open.
Decoupling at the connector pins: VBAT 10u + 100n, VCI 4.7u + 100n, IOVCC 1u + 100n, TP_VDD 1u + 100n.



Panel VBAT: external source or DK VDD, through the rev B load switch

J3 takes a PPK2 in source-meter mode for the 3.0–4.5 V sweep and the current-vs-content runs.
J4 selects VBAT_IN: 1–2 = J3 (external), 2–3 = DK VDD (3.0 V, below the pin table's 3.7 V minimum).
U1 is the TPS22917 rev B keeps (U21, renamed DISP_VBAT_EN): same CT and QOD wiring as rev A.
R1 (DNP) bypasses U1 to rule it out.



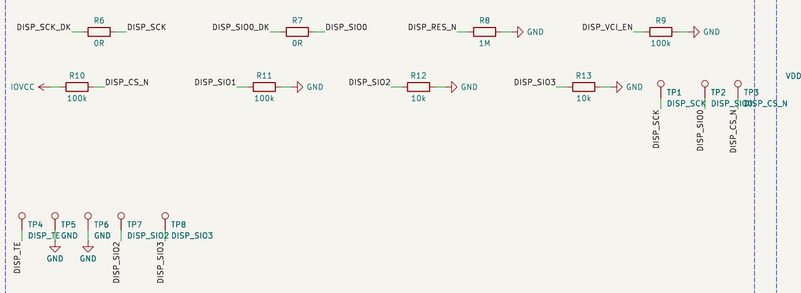
Panel logic rails from DK VDD, each separable

J5 shunt carries VCI + IOVCC + TP_VDD: remove it and put a PPK2 (ammeter) across to log the logic current.
R3–R5 split the three rails so any one can be fed or measured on its own.



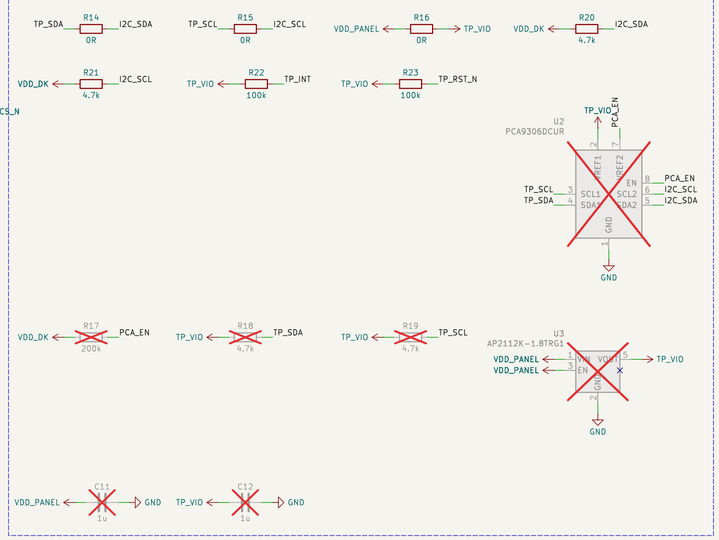
Display control and the 32 MHz lines

R6/R7 0R at the header: swap for 22–33R if the SCK/SIO0 edges ring (CO5300 needs <= 3.5 ns 20–80 %).
TP1–TP4 + TP5/TP6 GND beside the connector for the scope: probe at the panel end, with a spring ground.
DISP_RES_N 1M pull-down and DISP_VCI_EN 100k pull-down hold the panel off while the DK resets.
SIO1 (DCX) reads back on A4, a low-frequency pin: read ID/status at a reduced clock only.
SIO2/SIO3 are unused by the single-lane driver: 10k pull-downs, TP7/TP8 for a later QSPI test.



Touch I2C: direct at DK VDD, or level-shifted to 1.8 V

Fitted: R14/R15 0R tie the touch bus straight to P0.26/P0.27 at 3.0 V (Waveshare runs it at 3.3 V).
1.8 V option (the Huaxin/Viewe drawings): remove R14/R15/R16, fit U2, U3, C11, C12, R17, R18, R19.
In 1.8 V mode TP_INT and TP_RST_N are NOT shifted: TP_INT at 1.8 V is below the nRF's 0.7 VDD = 2.1 V VIH, and the nRF would drive TP_RST_N to 3.0 V. That mode answers 'does the bus work', not 'is it done'.



ERC power flags

Every rail here arrives from a header or through a jumper / OR link, which ERC cannot see as a source.



dial-me-spinner

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